PROJECT AND TEAM INFORMATION

Project Title

Console-Based Voting System

Student / Team Information

Team Name: **Team Code**	*CodeBallots* *DSCPP-III-2026-T286*
Team member 1 (Team Lead)	**Gupta Ruhee** Student ID: 2510379274 ruheegupta006@gmail.com 
Team member 2	**Duhan Anshul** Student ID: 2510010063 duhananshul.08@gmail.com 
Team member 3	**Thakur Kartik** Student ID: 2510360024 kartikthakur26776.dpskalyanpur@gmail.com 
Team member 4	**Tripathi Shreyash** Student ID: 2510012435 shreyashji777@gmail.com 

Under the guidance of Mr. Chirag Tyagi
Department of Computer Science
Graphic Era Deemed to be University
2 September 2026

PROPOSAL DESCRIPTION

Motivation

Paper-based voting works, but it becomes difficult to manage when the number of voters increases. Registration, checking voter details, counting votes, and preparing the final result all take time and leave room for mistakes. Our project aims to make this process simpler by creating a small voting system that can run directly on a computer. A voter can be registered, cast one vote, and then the result can be calculated by the program. We chose this problem because it gives us a practical way to apply the C++ concepts we are learning, especially classes, data structures, validation, and basic program design.

State of the Art / Current solution

In a traditional setup, voters are recorded on paper and votes are collected through physical ballots. After voting ends, the votes have to be counted and checked manually. This approach needs more time and careful supervision, and a mistake during counting can affect the result. Large electronic voting systems are available, but they normally require more infrastructure, security arrangements, databases, or specialised hardware. For this academic project, we are taking a simpler approach: a console application that keeps the required records in memory and focuses on the basic voting workflow rather than on a large-scale election environment.

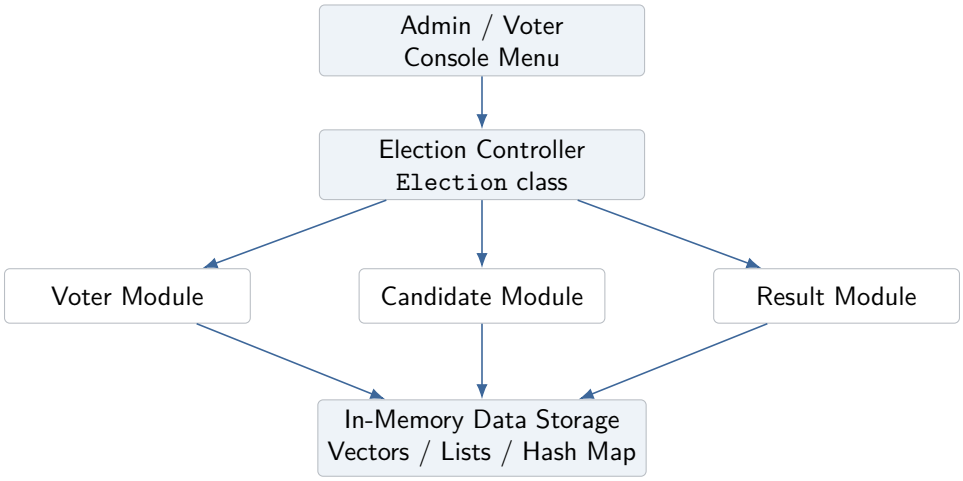
Project Goals and Milestones

The main goal is to build a working console-based voting program in C++. The system should allow candidates and voters to be registered, check voter IDs, accept a valid vote only once, and show the final result. We also want the program to have a clear structure so that each part can be understood and tested separately.
The planned milestones are: first, define the requirements and voting flow; second, design the classes for people, voters, candidates, and the election; third, add registration and validation; fourth, implement voting and duplicate-vote checking; fifth, add result calculation; and finally, test the program with valid, duplicate, and invalid inputs. In the future, the system can be improved by adding permanent data storage, a user-friendly graphical interface, stronger security, and network connectivity.

Project Approach

The system will be developed in C++ as a simple menu-based console application. The program will be organized using classes to keep the data and functions related to each part of the system together.The main classes will include Person, Voter, Candidate and Election. Encapsulation will be used to keep data protected while inheritance and polymorphism will be used when different classes need to share common features or behaviours. Candidate information can be stored using arrays or vectors. To quickly check voter IDs and make sure that a person cannot vote more than once a hash-based structure such, as 'unordered map' can be used. The program will rely on C++ libraries and will be tested using a regular C++11 or later development environment

System Architecture (High Level Diagram)



Project Outcome / Deliverables

The final deliverable will be a C++ console application that covers the complete basic voting process. It will include candidate registration, voter registration, voter-ID checking, one-vote-per-voter control, vote counting, and result declaration. The project will also show how OOP and data structures can be used together in a practical program. The expected result is a small, understandable, and portable system that gives the election organiser a quick way to manage votes and obtain the result without doing the counting manually.

Assumptions

The system is intended for a small-scale academic or local election and is assumed to run on one computer. Every registered voter has a unique voter ID, and the administrator controls candidate registration and the closing of the election. The current version stores information only while the program is running, so records are lost when the application closes. Biometric verification, OTP authentication, remote voting, and permanent database storage are outside the present scope.

References

1. Herbert Schildt, *C++: The Complete Reference*, McGraw-Hill Education.
2. E. Balagurusamy, *Object-Oriented Programming with C++*, McGraw-Hill Education.
3. Narasimha Karumanchi, *Data Structures and Algorithms Made Easy*, CareerMonk Publications.
4. C++ documentation: `https://cplusplus.com` and `https://en.cppreference.com`
5. GeeksforGeeks, data structures and OOP references: `https://www.geeksforgeeks.org`